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CIS 2266

Instructions

Iris Category Exercise

Using the iris dataset create a summary data plot for the following:

- Sepal length by Sepal width by species
- Petal length by Petal width by species

1. Load and Explore Data

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
filename = "iris.csv"
iris_df = pd.read_csv(filename)

# Display basic data overview
print("Summary statistics:")
display(iris_df.describe(include='all'))

# Check for data integrity (missing values)
print("Missing values by column:")
print(iris_df.isna().sum())

# Verify class distribution
print("\nSpecies distribution:")
iris_df["species"].value_counts()
```

Summary statistics:

	sepal_length	sepal_width	petal_length	petal_width	species
count	150.000000	150.000000	150.000000	150.000000	150
unique	NaN	NaN	NaN	NaN	3
top	NaN	NaN	NaN	NaN	setosa
freq	NaN	NaN	NaN	NaN	50
mean	5.843333	3.054000	3.758667	1.198667	NaN
std	0.828066	0.433594	1.764420	0.763161	NaN
min	4.300000	2.000000	1.000000	0.100000	NaN
25%	5.100000	2.800000	1.600000	0.300000	NaN
50%	5.800000	3.000000	4.350000	1.300000	NaN
75%	6.400000	3.300000	5.100000	1.800000	NaN
max	7.900000	4.400000	6.900000	2.500000	NaN

Missing values by column:

```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

Species distribution:

```
Out[1]: species
setosa      50
versicolor  50
virginica   50
Name: count, dtype: int64
```

2. Sepal Analysis (Manual Plotting)

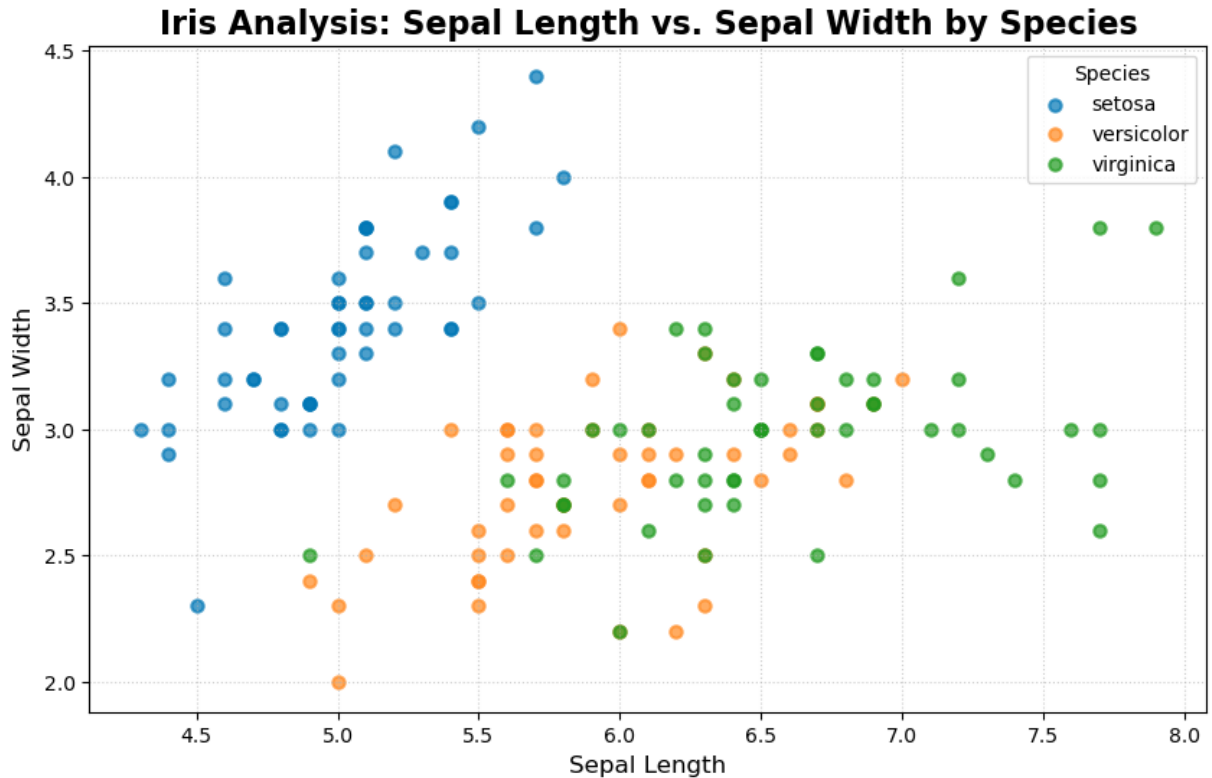
```
In [2]: plt.figure(figsize=(10, 6))

colors_dict = {'setosa': '#0077B6', 'versicolor': '#FF8920',
               'virginica': '#1F9A1F'}

# Plotting each species manually
for species, color in colors_dict.items():
    subset = iris_df[iris_df["species"] == species]
    plt.scatter(subset["sepal_length"], subset["sepal_width"],
                c=color, label=species, linewidth=1.5,
                marker = 'o', alpha=0.7)

# Formatting the plot
plt.title("Iris Analysis: Sepal Length vs. Sepal Width by Species",
          fontsize=16, fontweight='bold')
plt.xlabel("Sepal Length", fontsize=12)
```

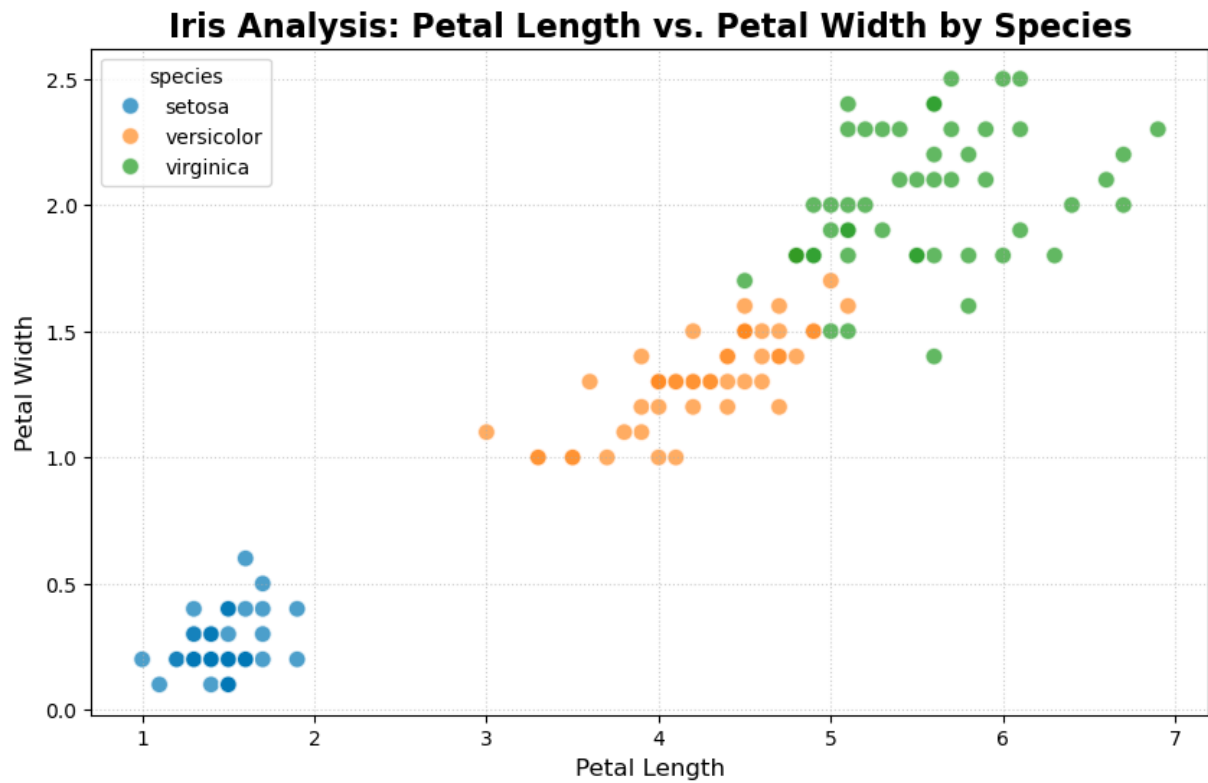
```
plt.ylabel("Sepal Width", fontsize=12)
plt.grid(True, linestyle=':', alpha = 0.6)
plt.legend(title='Species', fontsize=10)
plt.show()
```



3. Petal Analysis (Using Seaborn)

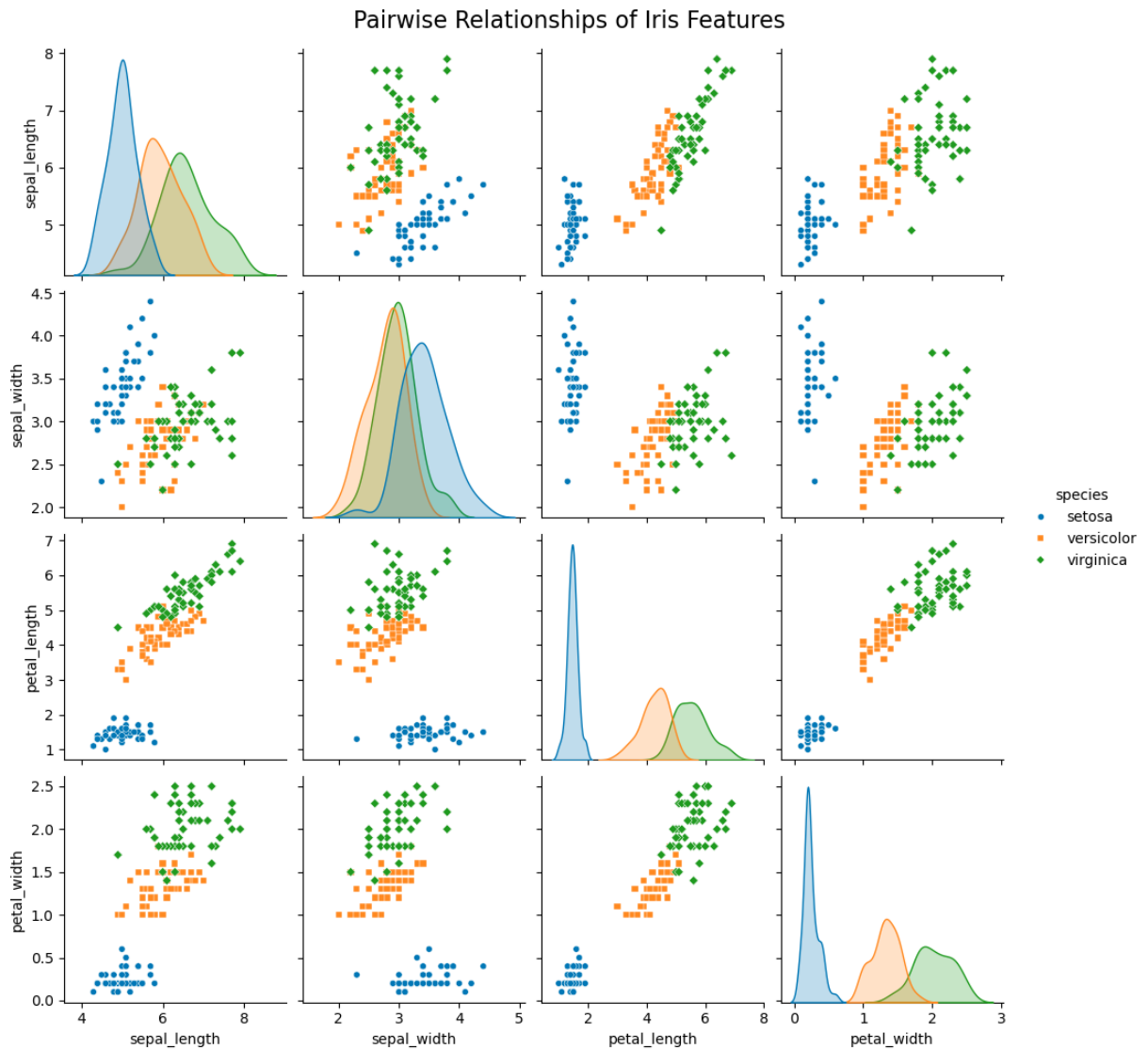
```
In [3]: plt.figure(figsize=(10, 6))
# Using Seaborn for more efficient categorical plotting
sns.scatterplot(data=iris_df, x='petal_length', y='petal_width',
                palette=colors_dict, hue='species', s=75,
                alpha=0.7, linewidth=1.0)

# Formatting the plot
plt.title("Iris Analysis: Petal Length vs. Petal Width by Species",
          fontsize=16, fontweight='bold')
plt.xlabel("Petal Length", fontsize=12)
plt.ylabel("Petal Width", fontsize=12)
plt.grid(True, linestyle=':', alpha = 0.6)
plt.show()
```



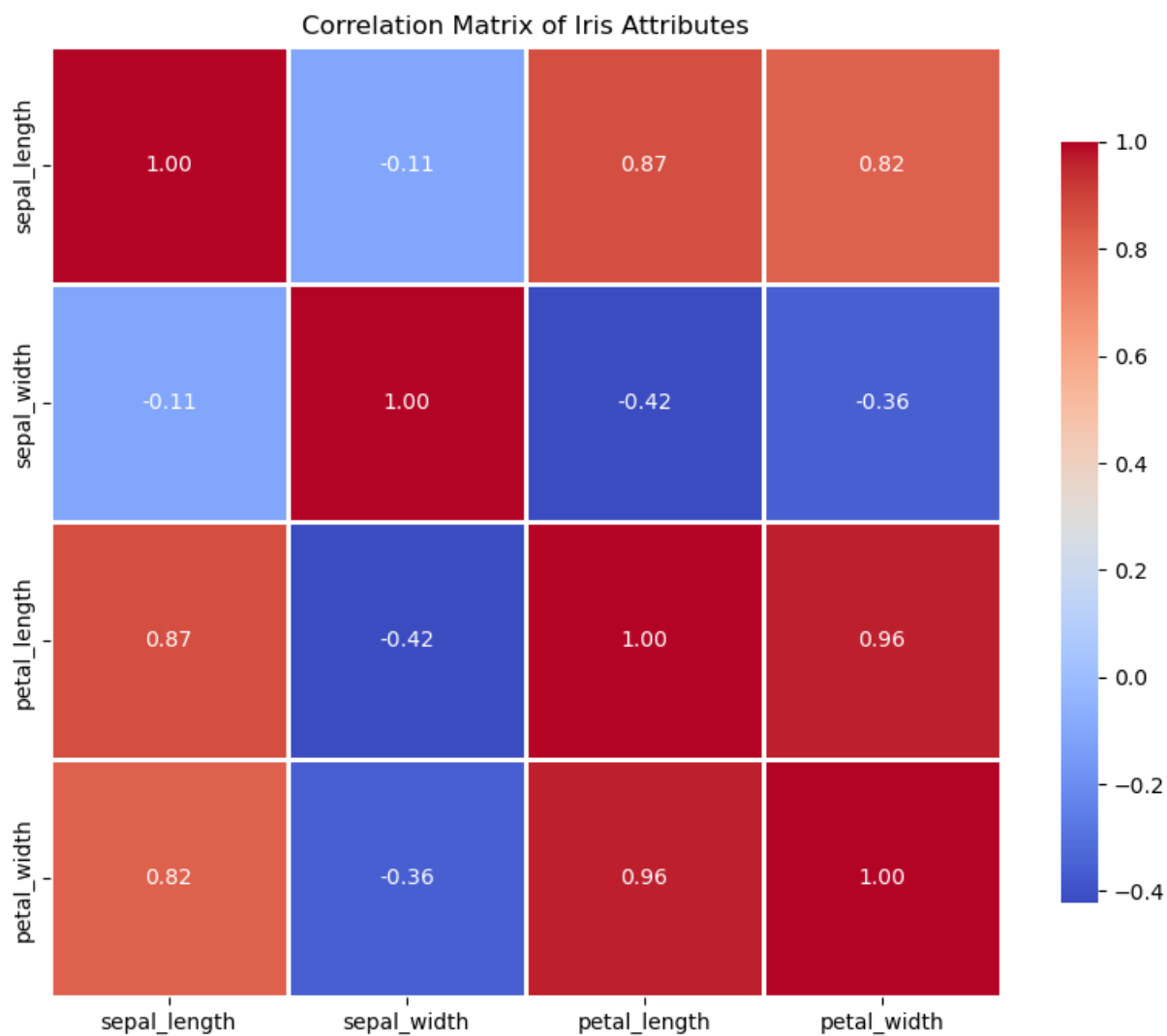
4. Visualizing all feature combinations to identify key differentiators (Pairplot)

```
In [4]: sns.pairplot(iris_df, hue='species', palette=colors_dict,
                    markers=['o', 's', 'D'], plot_kws={'s': 20})
plt.suptitle("Pairwise Relationships of Iris Features",
             y=1.02, fontsize=16)
plt.show()
```



5. Feature Correlation: Identifying mathematical relationships between physical attributes

```
In [5]: plt.figure(figsize=(10,8))
corr = iris_df.drop('species', axis=1).corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f',
            linewidths=1, square=True, cbar_kws={"shrink": .8})
plt.title("Correlation Matrix of Iris Attributes")
plt.show()
```



In []: